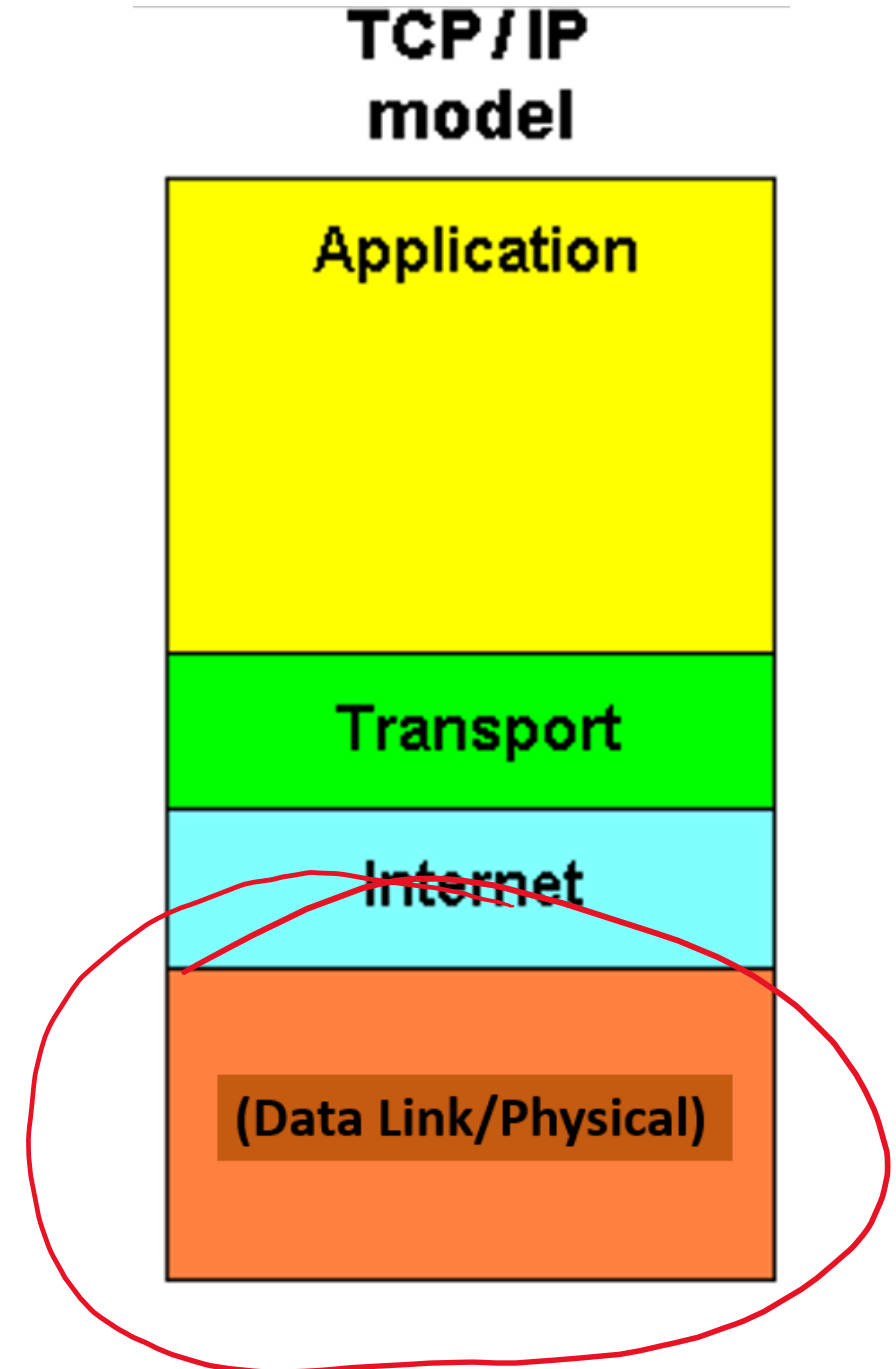


Physical/Data Link Layer



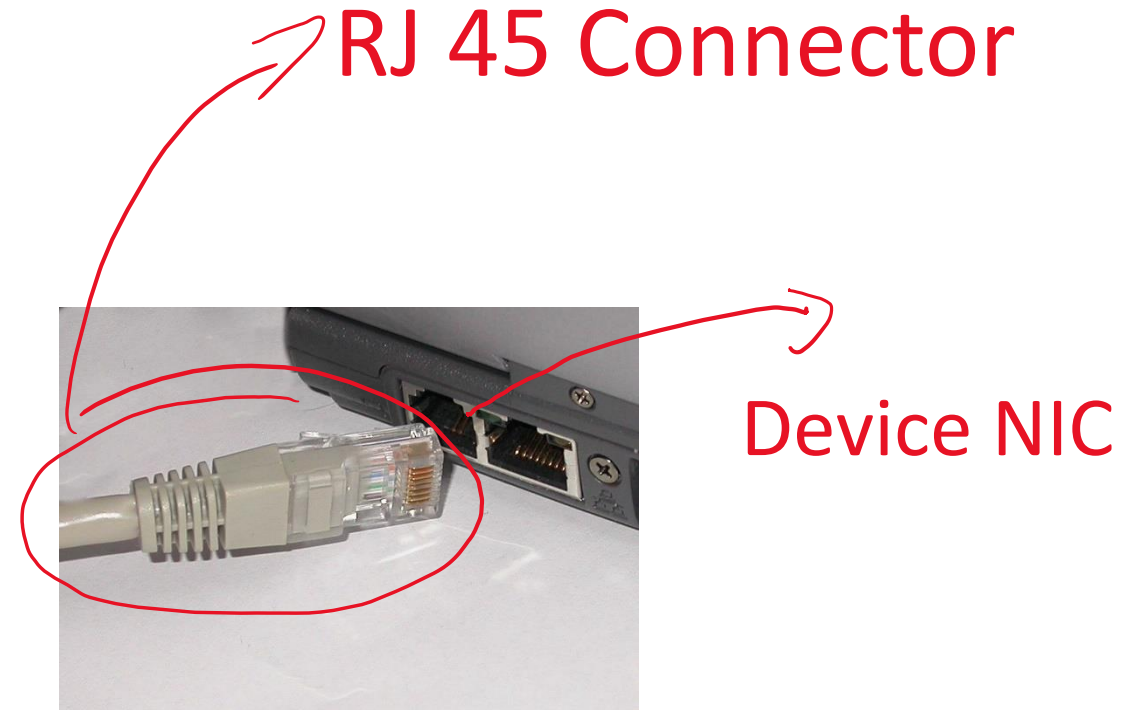
Agenda (1st Part)

- Physical Media
- Purpose of Data Link Layer
- More about topologies
- Data Link Frame
- Ethernet



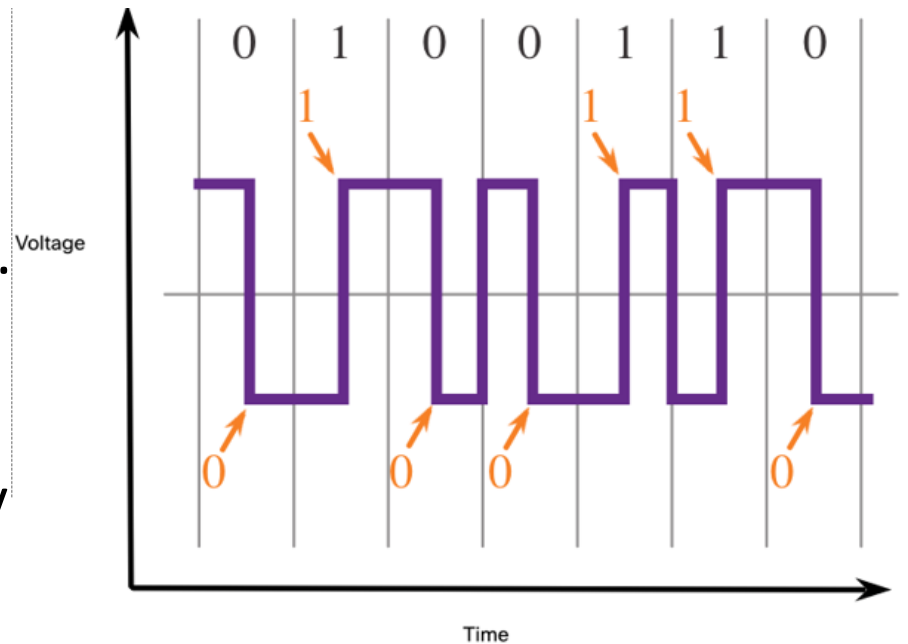
1. Physical Media

- Networks inherently physical
 - Must have a Physical Connection
- A Network Interface Card (NIC)/Network Adapter connects a device to the network.
- Some devices may contain more than 1 NIC (How many does your RPi have?)
- Can have different performance levels.



1. Physical Media Characteristics

- NICs/cables are specified in standards associated with the physical layer.
- Signaling:
 - the method (voltage, current, RF, light) used to represent binary values on the wire.
- Encoding is how the 1s and 0s will be represented.
 - Manchester encoding: A 0 is represented by a high to low voltage transition and a 1 is represented as a low to high voltage transition.



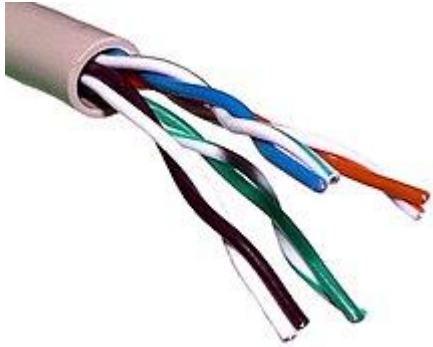
1. Physical Media: Bandwidth, Throughput, Latency

of Bandwidth	Abbreviation	Equivalence
er second	bps	1 bps = fundamental unit of bandwidth
s per second	Kbps	1 Kbps = 1,000 bps = 10^3 bps
bits per second	Mbps	1 Mbps = 1,000,000 bps = 10^6 bps
its per second	Gbps	1 Gbps = 1,000,000,000 bps = 10^9 bps
ts per second	Tbps	1 Tbps = 1,000,000,000,000 bps = 10^{12} bps

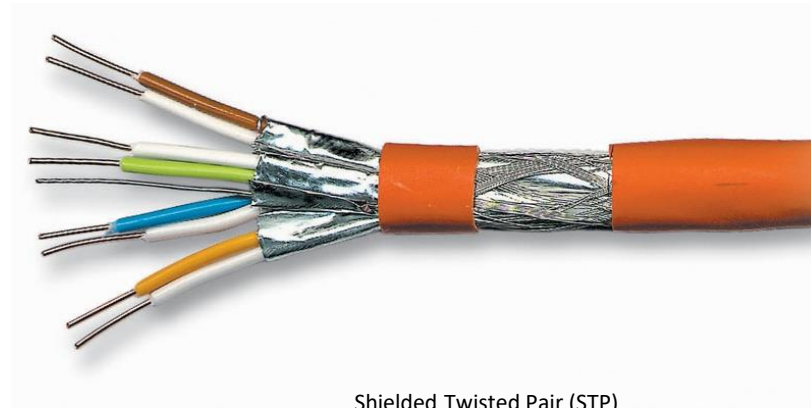
“Never underestimate the bandwidth of a station wagon full of tapes hurtling down the highway” – Andrew Tanenbaum, 1981.

- Bandwidth is the max data carrying capacity
- Remember last lab: how many bits can be transmitted in a second.
- Terminology:
 - Latency: The time, including delays, for data to travel from one point to another.
 - Throughput: actual measure of transfer of bits across media in given time.
 - Throughput is lower than Bandwidth
- Sending data via FedEx, theoretically, has better bandwidth than the internet
 - However, 80,000,000-millisecond ping times (<https://what-if.xkcd.com/31/>)

1. Physical Media: Copper Cabling



Unshielded Twisted-Pair (UTP)



Shielded Twisted Pair (STP)



Coaxial Cable

2. Data Link Layer: Main Purpose

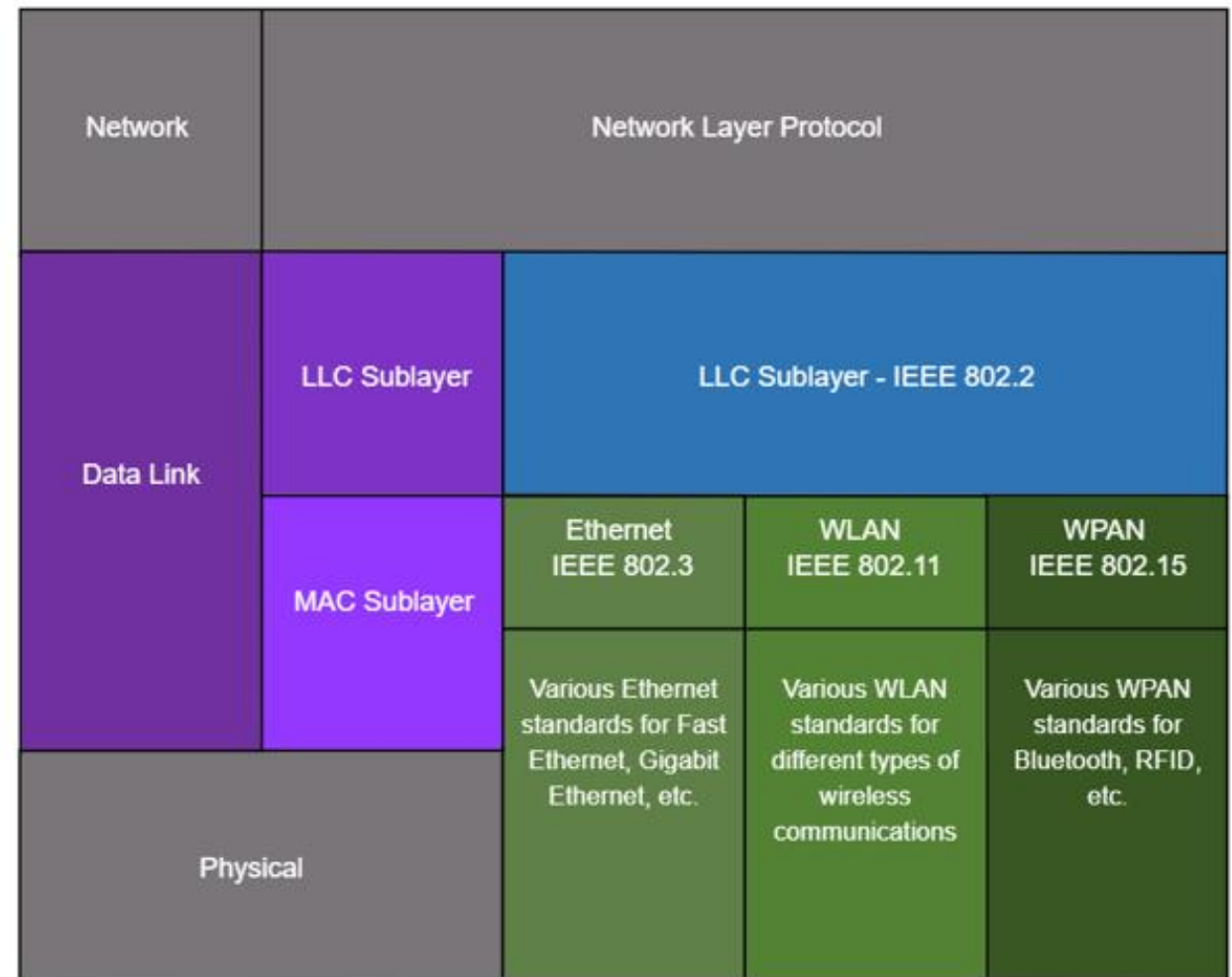
RESPONSIBLE FOR
COMMUNICATION
BETWEEN END DEVICE
NETWORK INTERFACE
CARDS(NIC)

PROVIDES ACCESS TO
PHYSICAL MEDIA FOR
UPPER LAYER PROTOCOLS
(I.E. NETWORK LAYER IN
TCP/IP)

ERROR DETECTION AND
FRAME REJECTION USING

2. Data Link Layer: Standards

- The Data Link Layer has two sublayers. **Logical Link Control (LLC)** and **Media Access Control (MAC)**.
 - The LLC sublayer communicates between the networking software at the upper layers and the device hardware at the lower layers.
 - The MAC sublayer is responsible for data encapsulation and media access control.



2. Data Link Layer: Access to Physical Media

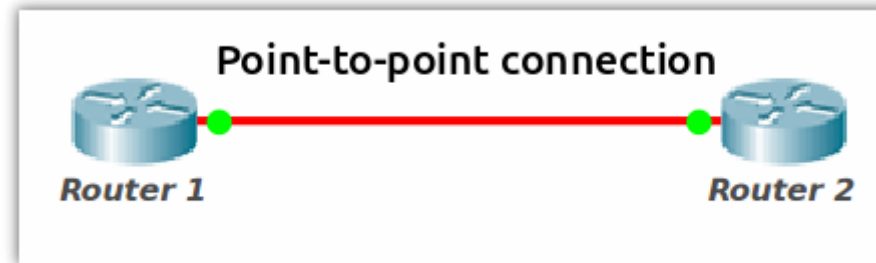
- Communication between nodes might experience many Data Link Layers and Media Types
 - For example, Wifi to Ethernet
- Each “hop”, a router will carry out the following functions:
 1. Accepts a frame from the network medium
 2. De-encapsulates the frame to expose the encapsulated packet
 3. Re-encapsulates the packet into a new frame
 4. Forwards the new frame on the medium of the next network segment

2. Data Link Layer: Topologies

- Remember, Network Topology is the arrangement and relationship of nodes and the interconnection between them:
- **Physical Topology:** illustrates physical connections
- **Logical topology:** virtual connections and specifies interfaces and addressing schemes

2. Data Link Layer: WAN Topologies

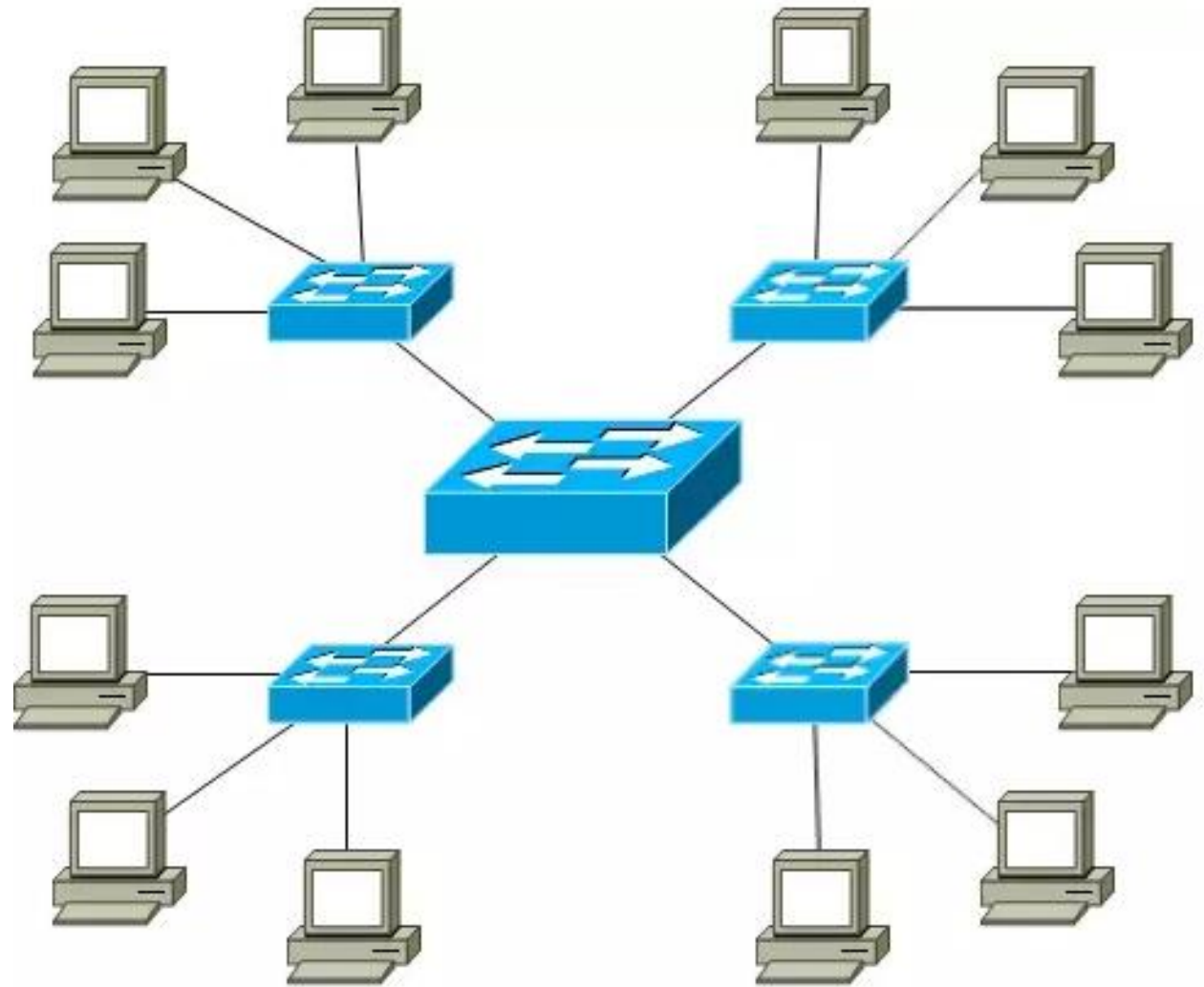
- Common WAN topologies are: Point to Point, Hub and Spoke and Mesh
- Point to point:
 - directly connect two nodes
 - nodes may not share the media with other hosts (means P2P protocols are simple)



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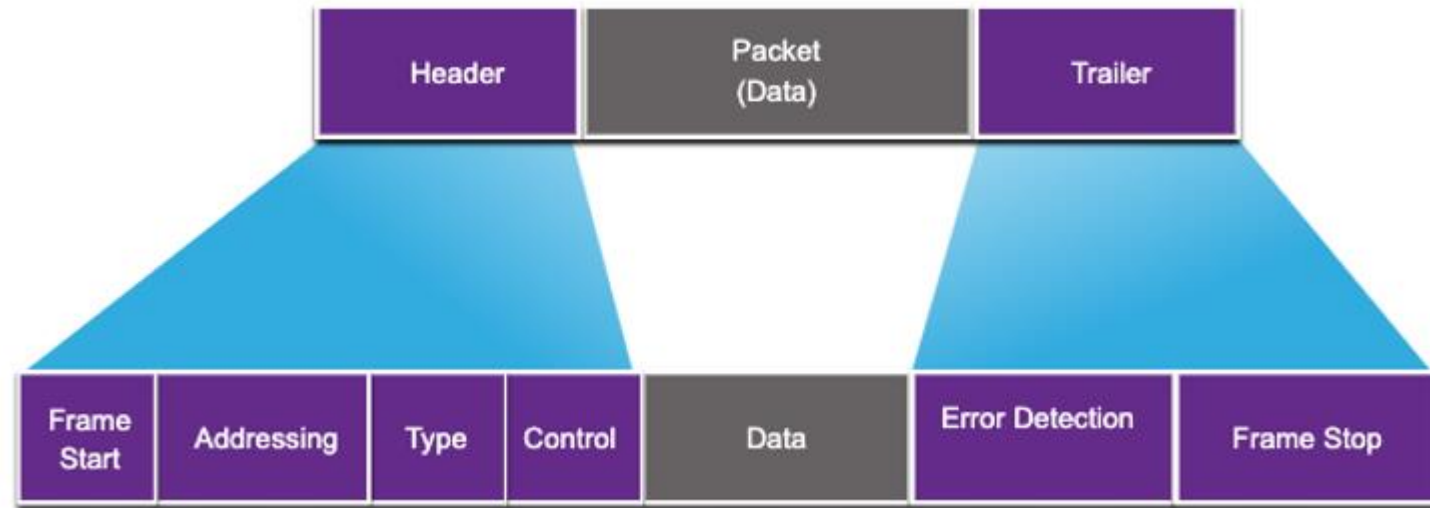
2. Data Link Layer: LAN Topologies

- End devices on LANs typically use star/extended star topology.
 - Generally easy to scale and troubleshoot.



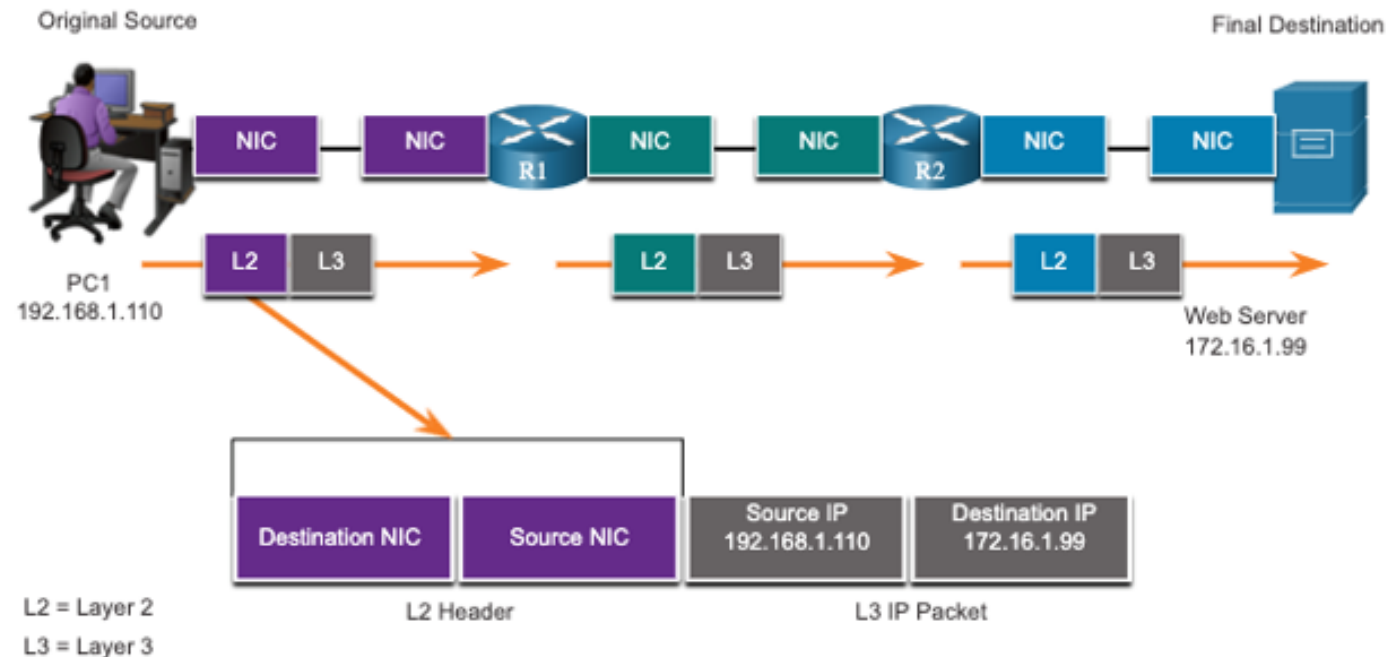
Data Link Frame

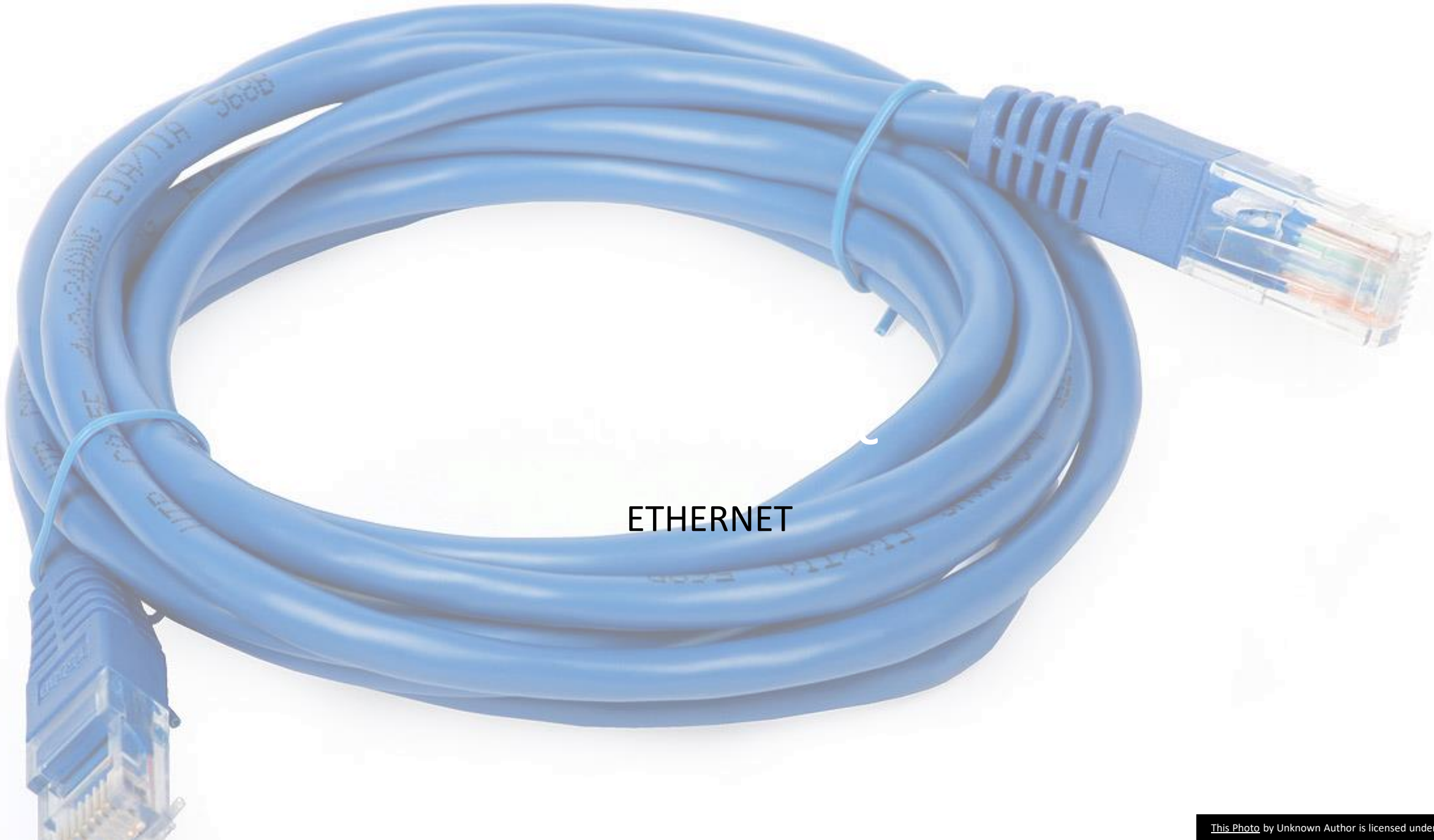
Data Link Layer: Frame Fields:



Data Link Layer: Addresses

- Also referred to as a physical address or MAC address.
- Contained in the frame header.
- **Used only for local delivery of a frame on the link.**
- **Updated by each device that forwards the frame.**

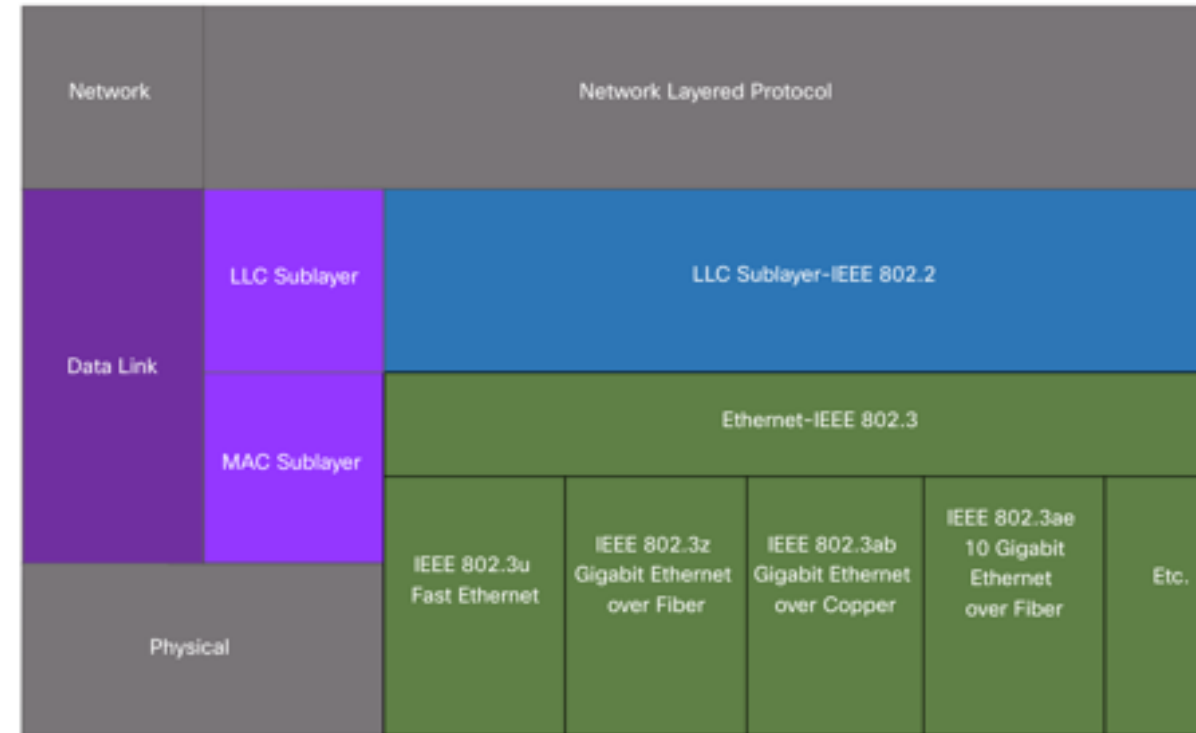




ETHERNET

Ethernet: Characteristics

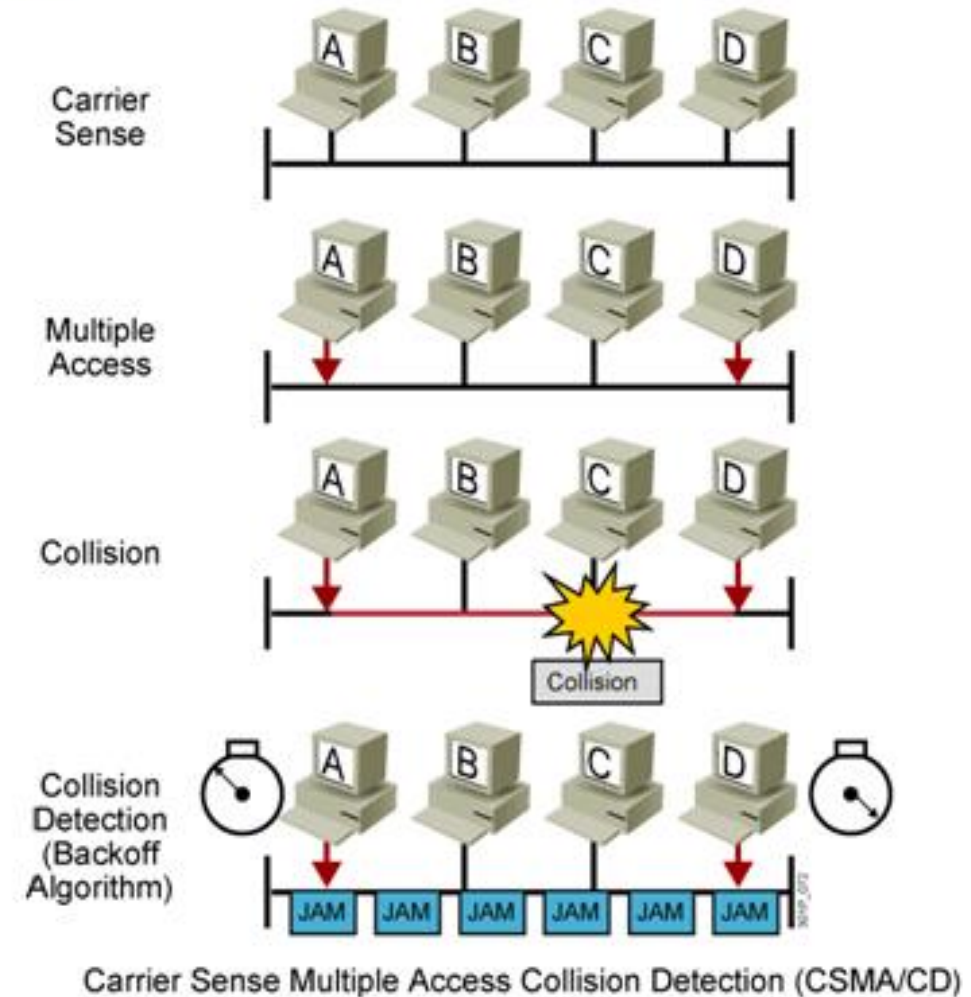
- Operates across Data Link/physical layer
- Provides the following that pure physical layers do not:
 - Connects to upper layers (i.e. network)
 - Provides mechanism to recognise devices
 - Organises bits into frames
- Provides encapsulation into “Frames”
- Ethernet Provides Media Access Control
 - Placement and removal of frames onto media
 - Media access control for ethernet is CSMA/CD
 - All devices on network segment share media
 - All devices receive all frames transmitted on network

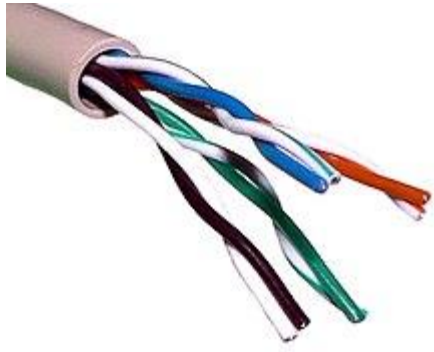


Ethernet: CSMA/CD

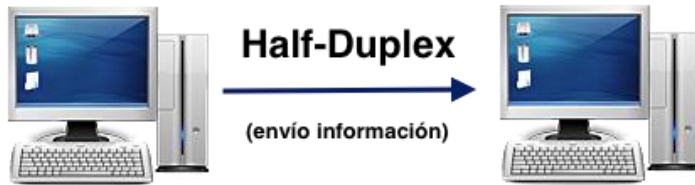
CSMA/CD

- Some older Ethernet networks use a bus topology or hubs. In shared medium, uses a contention-based access method, carrier sense multiple access/collision detection (CSMA/CD).
- Ethernet LANs of today use switches that operate in full-duplex. Full-duplex communications with Ethernet switches do not require access control through CSMA/CD.





Ethernet Collision Detection



- In shared media, only one device can transmit
- More devices on network => more collisions
- Switches reduce collisions
- Switches isolate each port and can send a frame to its destination (if known) rather than every device
- Often uses twisted pair cabling:
 - Allows for one pair for transmission, one for receiving.
- The capability to do both simultaneously is called full duplex
 - No contention for media => no collision domain
=> No need for CSMA/CD